

Single-Cell RNA-seq with BioLang

[↺ Reset Interpreter](#)

38 code blocks • state persists between runs • use ▶▶ to auto-run dependencies

A tissue is not one thing. A blood sample contains many immune populations. A tumor contains malignant cells, immune cells, blood-vessel cells, connective tissue cells, and damaged cells. Bulk RNA sequencing mixes their RNA into one average. Single-cell RNA sequencing (scRNA-seq) measures thousands of those cells separately.

That extra detail is useful, but it creates a new problem: a colorful map can be easy to produce and easy to overinterpret. This book teaches both the workflow and the reasoning that makes the workflow defensible.

Who this book is for

- **Biologists** can connect laboratory decisions to computational artifacts.
- **Researchers** can turn a biological question into a reproducible analysis.
- **Programmers** can understand the data model, algorithms, and performance boundaries without first becoming molecular biologists.
- **Clinicians** can understand what a single-cell result does and does not say about a patient or treatment.
- **Students and interested readers** can begin without prior RNA-seq knowledge.

Each chapter answers five questions:

1. **What** is this concept?
2. **Why** does it matter?
3. **How** is it represented or analyzed?
4. **When and where** should it be used?
5. **What can fool us?**

What you will build

You will analyze a deterministic, tumor-like 10x count matrix with four known cell populations. The fixture is intentionally small enough for a laptop and is generated locally rather than stored in the repository. You will:

- inspect raw counts and quality metrics;
- remove implausible cells and rarely detected genes;
- normalize counts and select variable genes;
- compute PCA, a neighbor graph, Leiden clusters, and UMAP;
- inspect markers and propose cell-type labels;
- compare BioLang clusters with Scanpy and Seurat;
- plan a replicate-aware multi-sample analysis;
- record the decisions needed to reproduce the result.

A necessary boundary

This book teaches research analysis. A cluster, marker, or association is not a clinical diagnosis. Clinical use requires an independently validated assay, quality system, predefined decision rule, appropriate population, regulatory review, and human oversight.

BioLang's `singlecell` package is useful for transparent preprocessing, exploration, and reproducible workflows. It does not make biological judgment automatic. The analyst remains responsible for study design, metadata, cell annotations, statistical units, and interpretation.

How to read it

Read Parts I and II in order on a first pass. Part III is organized by scientific question. Part IV should be read before publishing or handing a result to a collaborator. Terms in **bold** are collected in the glossary, and every package call is summarized in the API appendix.

One Tissue, Many Cells

The everyday analogy

Suppose a hospital asks whether patients are waiting too long. One number for the whole hospital hides the difference between emergency, radiology, pharmacy, and reception. Measuring each department separately reveals where the delay occurs.

Bulk RNA-seq is the hospital-wide average. Single-cell RNA-seq is closer to measuring each room separately. It can reveal a rare population, a changing cell state, or a different mixture of cells that the average hides.

What RNA tells us

DNA is a long-term instruction store. A **gene** is a named region that can contribute to a biological product. When a gene is active, the cell can make RNA copies called **transcripts**. Messenger RNA is not a direct measurement of protein, behavior, or disease, but its abundance is evidence about what a cell was doing when it was captured.

Single-cell RNA-seq asks: which RNA molecules were observed in each captured cell or nucleus?

The usual result is a matrix:

	Gene A	Gene B	Gene C
Cell 1	0	4	1
Cell 2	0	5	0
Cell 3	7	0	2

Rows are cells, columns are genes, and entries are observed molecule counts. Most entries are zero, so the matrix is **sparse**.

Questions it can answer

Single-cell analysis is useful when the variation among cells matters:

- Which cell populations are present in a tumor?
- Does treatment change a cell type's abundance or state?
- Which immune population expresses an inflammatory program?
- What intermediate states appear during development?
- Which cells respond to infection?

It is less useful when the scientific question concerns only a well-purified, uniform population or when sample replication is too weak to support the comparison.

Four viewpoints

The biologist asks whether the populations and markers make biological sense.

The computational researcher asks whether processing choices created the pattern. **The programmer** asks whether data structures and algorithms scale.

The clinician asks whether the finding was replicated and whether it changes a validated decision.

All four viewpoints are needed. A technically correct cluster can still be a doublet. A plausible marker can still be caused by batch. A statistically significant gene can still be clinically irrelevant.

Checkpoint

Before continuing, explain this sentence in your own words:

Single-cell RNA-seq measures noisy evidence about cell state, not a complete inventory of everything a cell contains or does.

What the Machine Measures

From tissue to counts

A common droplet workflow follows this path:

1. A tissue or blood sample is separated into cells or nuclei.
2. Individual cells enter droplets with barcoded beads.
3. RNA is copied into DNA while receiving a **cell barcode** and a **unique molecular identifier** (UMI).
4. The DNA is sequenced.
5. Reads are aligned or assigned to genes.
6. Reads with the same cell barcode, gene, and corrected UMI are collapsed into one observed molecule count.

10x Genomics describes Cell Ranger as performing alignment, filtering, barcode processing, and UMI counting to produce a feature-barcode matrix. A UMI is used to reduce the effect of sequencing the same captured molecule many times. See the Cell Ranger overview and gene-expression algorithm.

Barcode is not identity

A barcode identifies a captured droplet, not a known biological cell type. Several complications follow:

- An empty droplet can contain ambient RNA.
- One droplet can capture two or more cells, producing a **doublet** or **multiplet**.
- A damaged cell can lose cytoplasmic RNA.
- Two samples can use overlapping barcode sequences.
- A nucleus and a whole cell measure different RNA compartments.

The count matrix therefore begins as a set of candidate cellular libraries, not perfect cells.

Why so many zeros?

A zero can mean at least three things:

1. the gene was inactive;
2. the transcript existed but was not captured;
3. the transcript was captured but not confidently assigned.

This is why absence of a marker in one cell is weak evidence. Patterns across many genes and cells are more reliable.

The 10x MEX directory

BioLang reads the common Matrix Exchange layout:

```
filtered_feature_bc_matrix/  
  matrix.mtx.gz  
  features.tsv.gz  
  barcodes.tsv.gz
```

`matrix.mtx.gz` stores only nonzero entries. `features.tsv.gz` names genes or other features. `barcodes.tsv.gz` names cell barcodes. The file matrix is typically genes by cells; `sc.load()` exposes it as cells by genes.

Raw, filtered, and normalized data

- **Raw counts** are observed UMI counts before downstream transformation.
- A **filtered matrix** contains barcodes selected by a cell-calling process.
- **Normalized values** make cells with different library sizes more comparable for exploration.
- **Scaled values** center and standardize genes for some algorithms.

Never replace raw counts without recording what happened. Count-based statistical models may need them later.

Checkpoint

For every matrix you receive, ask: Is it raw or normalized? Are rows genes or cells? How were cells called? Are gene symbols unique? Which genome annotation was used? Which sample produced each barcode?

Design the Study Before the Analysis

The experimental unit

If three patients contribute 5,000 cells each, the treatment comparison usually has three biological replicates per group, not 15,000 independent replicates. Cells from the same person share genetics, environment, collection, and processing. Treating them as independent is **pseudoreplication**.

This distinction changes the question:

- “Which genes distinguish these two clusters in this dataset?” is exploratory.
- “Which genes respond to treatment across patients?” requires patient-level replication and a model that accounts for it.

Start with a question table

Write this before opening the matrix:

Field	Example
Population	Adults with treatment-resistant disease
Comparison	Drug vs matched baseline
Biological unit	Patient
Tissue	Peripheral blood
Primary outcome	State change within CD8 T cells
Main confounders	Patient, processing day, sex, disease severity
Exclusions	Predefined low viability or failed library
Validation	Independent cohort and orthogonal assay

The table prevents a visually interesting result from silently replacing the original question.

Balance biology and processing

If all controls are processed Monday and all treated samples Tuesday, treatment and processing day are **confounded**. No algorithm can prove which caused the difference. Randomize or balance samples across processing batches where possible. Record donor, condition, collection time, chemistry, lane, operator, and reference version.

Power is not only cell count

More cells improve the description of within-sample diversity and help detect rare populations. More independent samples improve inference about a population. Ten thousand cells from one donor do not substitute for additional donors when the claim concerns people.

Clinical and ethical context

Single-cell data can contain germline variants, disease information, and rare cell states that increase re-identification risk. Keep identifiers separate, apply appropriate access controls, record consent limitations, and avoid publishing unrestricted cell-level metadata that can expose a participant.

For clinical questions, define in advance whether the result is discovery, validation, or decision support. Do not move between those roles without new evidence and governance.

Analysis contract

Create a short analysis contract:

```
Question: Does treatment alter the inflammatory state of monocytes?
Unit: patient
Samples: 8 baseline, 8 treated, paired where available
Cell annotation: blinded marker review plus reference mapping
Primary comparison: monocyte pseudobulk counts, paired design
Sensitivity: alternate QC thresholds and cluster resolutions
Validation: held-out cohort plus flow cytometry panel
```

This plain-language contract is more valuable than a long script with no stated claim.

Set Up BioLang and the Data

Install and verify

Install a current BioLang release, then confirm the CLI and capabilities:

```
bl version
bl doctor
```

From the BioLang repository, build the development binary with:

```
cargo build -p bl-cli
```

Copy the examples from an installed package into a standalone working directory:

```
bl examples singlecell --copy singlecell-examples
cd singlecell-examples
```

Package authors working from a BioLang source checkout can use `bl examples packages/singlecell --copy singlecell-examples` instead. The copied directory includes the BioLang, notebook, Python, and validation example files; it does not require the full BioLang repository.

Generate the teaching matrix

The fixture contains four populations with distinct marker blocks, shared background genes, mitochondrial genes, and low-information droplets. It is synthetic: no person or patient data is involved.

```
python make_demo_10x.py --output nsclc_like
```

`--output` is relative to the current directory. With the commands above, the fixture is created at:


```
<your working directory>/nsc_lcl_like/
```

It contains `matrix.mtx.gz`, `features.tsv.gz`, `barcodes.tsv.gz`, and `truth.csv`. The later `sc.load("nsc_lcl_like")` calls resolve that same directory because the examples are run from the copied working directory.

Expected summary:

```
168 genes x 265 barcodes (25 junk)
output: <your working directory>/nsc_lcl_like
```

The generator uses a fixed seed. Re-running it produces the same logical dataset, making it suitable for tests and comparisons.

Load it

Requires CLI: package imports and local filesystem access are not available in the browser runner.

#1

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let cells = sc.load("nsc_lcl_like")
println(sc.summary(cells))
```

The object is a BioLang record. Important fields are:

Field	Meaning
<code>matrix</code>	Raw cells-by-genes count matrix
<code>layers.counts</code>	Preserved raw counts
<code>genes</code>	Gene names in matrix order
<code>barcodes</code>	Cell barcodes in matrix order
<code>obs</code>	Cell metadata table
<code>var</code>	Gene metadata table

Field	Meaning
<code>n_cells</code> , <code>n_genes</code>	Current dimensions

The count matrix remains sparse after 10x loading, filtering, normalization, and HVG selection. Compact PCA scores are dense because every cell has a score on every retained component.

Use an in-memory matrix

For a tiny test, construct cells directly:

Requires CLI: this example imports the `singlecell` package.

#2

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let counts = matrix([
  [8, 0, 1, 0],
  [7, 0, 2, 0],
  [0, 9, 0, 1],
  [0, 8, 0, 2]
])
let tiny = sc.from_matrix(
  counts,
  ["T_MARKER", "B_MARKER", "HOUSEKEEPING", "MT-ND1"],
  ["cell_1", "cell_2", "cell_3", "cell_4"]
)
println(sc.summary(tiny))
```

Keep source and generated data separate

A reproducible project can use:

```
project/  
  README.md  
  biolang.toml  
  scripts/  
  data/raw/          # immutable or checksummed inputs  
  data/derived/      # generated matrices  
  results/tables/  
  results/figures/  
  validation/
```

Do not commit private or very large matrices merely to make a script look self-contained. Record a stable accession or an approved storage location, checksum the input, and provide a deterministic download or generation step.

Quality Control

Why QC exists

Quality control asks whether each barcode is a plausible measurement of one useful cell. It is not a ritual for applying universal thresholds. Tissue, protocol, chemistry, and biological state all change the distributions.

Three common per-cell metrics are:

- `total_counts` : total observed UMIs;
- `n_genes` : number of genes with a nonzero count;
- `pct_mito` : percentage of counts assigned to mitochondrial genes.

Low totals or few detected genes can indicate empty droplets or damaged cells. Very high totals or gene counts can indicate doublets. High mitochondrial fraction can indicate stress or membrane damage, but it can also be normal in some tissues.

Inspect before filtering

Requires CLI: this example imports the package and reads local files.

#3

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("nsclc_like") |> sc.qc()
println(head(raw.cell_qc_table, 8))
println(head(raw.gene_qc_table, 8))
```

Plot or summarize each metric by sample. A global threshold can unfairly remove one sample when library depth differs.

Filter with stated reasons

The teaching matrix has only 168 genes, so its minimum is intentionally much lower than a real whole-transcriptome matrix:

Requires CLI: this example imports the package and reads local files.

#4

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("nsclc_like")
let clean = raw
  |> sc.filter_genes(3)
  |> sc.filter_cells(20, 2500, 5.0)

println("before: " + str(raw.n_cells))
println("after:  " + str(clean.n_cells))
```

For a real matrix with roughly 20,000 to 35,000 genes, a starting exploratory range might be 200 to 5,000 detected genes and a tissue-appropriate mitochondrial threshold. These are starting points, not standards.

Empty droplets and ambient RNA

A filtered Cell Ranger matrix has already undergone cell calling, but it can still contain borderline barcodes. Conversely, aggressive filtering can discard small, low-RNA cell types. The EmptyDrops paper formalized a test against the ambient RNA profile and showed why total-count thresholds alone can miss biologically meaningful cells.

BioLang's current package provides metric-based filtering, not an EmptyDrops implementation. If raw droplet cell calling is scientifically important, run a validated cell-calling method upstream and record it.

Doublets

`sc.doublets()` simulates mixtures and assigns a score; `sc.flag_doublets()` applies a threshold:

Requires CLI: this example imports the package.

#5

▶ CLI Only

Requires CLI — run with: bl run script.bl

```
import "singlecell" as sc

let scored = clean |> sc.normalize() |> sc.variable_genes(100) |>
sc.doublets(500)
let flagged = scored |> sc.flag_doublets(0.5)
println(flagged.is_doublet |> filter(|x| x) |> len)
```

A computational flag is evidence, not certainty. The Scrublet study describes how simulated doublets and nearest neighbors can identify hybrid profiles.

What can fool you

- Mitochondrial gene recognition depends on gene symbols such as `MT-...`.
- Removing all high-count cells can erase a genuinely large cell type.
- Filtering samples separately and then merging is often safer than one global cutoff.
- A low-quality cluster should not automatically be relabeled as a novel type.
- Report cells before and after every filter, per sample.

Normalize and Select Signal

Why library size matters

One cell may have 20,000 observed UMIs and another 5,000. Directly comparing raw counts makes deeply sampled cells look globally more active. A common exploratory transformation divides each cell by its total, multiplies by a target such as 10,000, and applies $\log(1 + x)$.

Requires CLI: this example imports the package.

#6

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("nsc_lc_like")
|> sc.filter_genes(3)
|> sc.filter_cells(20, 2500, 5.0)
|> sc.normalize(10000.0)

println(sc.summary(obj))
```

Raw counts remain in `layers.counts`; normalized values are in `norm_matrix` and `layers.lognorm`.

What normalization does not do

Normalization does not:

- recover transcripts that were not captured;
- remove batch effects;
- prove that a gene is differentially expressed;
- make every cell biologically comparable;
- replace a replicate-aware statistical model.

It creates a more useful representation for exploration and many downstream geometric methods.

Highly variable genes

Thousands of genes vary little or mostly add noise. Highly variable gene (HVG) selection retains a subset whose variation can help describe cell-to-cell structure:

Requires CLI: this example imports the package.

#7

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("nsclc_like")
|> sc.filter_genes(3)
|> sc.filter_cells(20, 2500, 5.0)
|> sc.normalize()
|> sc.variable_genes(100)

println(take(obj.hvg_genes, 12))
```

BioLang currently ranks genes using dispersion (coefficient of variation squared) across all genes at once. Scanpy's `seurat` flavor and Seurat's VST first bin genes by mean expression and rank dispersion within each bin, which compensates for the fact that low-mean genes have higher dispersion for purely sampling reasons. BioLang's unbinned ranking therefore leans further toward low-expression genes. Exact HVG lists are not expected to match across tools; compare downstream structure and marker behavior, not gene-list identity.

SCTransform

`sc.sctransform()` provides regularized negative-binomial residuals as an alternative transformation:

Requires CLI: this example imports the package.

#8

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let transformed = sc.load("nsclc_like")
|> sc.filter_genes(3)
|> sc.filter_cells(20, 2500, 5.0)
|> sc.sctransform()

println(sc.summary(transformed))
```

Note that the result is dense. Pearson residuals are nonzero where counts were zero, so unlike `normalize()` this does not preserve sparsity — check the cell and gene counts before running it on a large matrix.

Do not mix normalization strategies without a reason. Record which matrix each later method used.

Decision record

For this step, save:

- the normalization method and target;
- whether raw counts were preserved;
- the HVG method and number selected;
- genes deliberately excluded, such as mitochondrial or ribosomal sets;
- plots or summaries used to justify the choice.

PCA, Neighbors, and Clusters

From thousands of genes to a map

Each cell begins as a point in a space with one dimension per gene. This space is noisy and difficult to inspect. The standard workflow builds progressively simpler representations:

1. **PCA** finds directions that explain large patterns of variation.
2. A **k-nearest-neighbor graph** connects similar cells in PCA space.
3. **Leiden clustering** partitions the graph into connected communities.
4. **UMAP** produces a two-dimensional visualization.

These are related but distinct. UMAP coordinates are for visualization; Leiden uses the neighbor graph. Clusters should not be defined by drawing shapes on the UMAP.

Build the structure

Requires CLI: this example imports the package and reads local files.

#9

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.load("nsc_lc_like")
  |> sc.filter_genes(3)
  |> sc.filter_cells(20, 2500, 5.0)
  |> sc.normalize()
  |> sc.variable_genes(100)
  |> sc.run_pca(20)
  |> sc.neighbors(15)
  |> sc.cluster_leiden(15, 0.5)

println("clusters: " + str(obj.clusters |> unique |> len))
sc.plot_elbow(obj, 15)
```

BioLang's sparse PCA centers features mathematically without materializing a dense centered expression matrix. The compact cell-by-component scores are dense.

Parameters are scientific decisions

`n_pcs` controls how much structure reaches the graph. `k` controls neighborhood scale. `resolution` controls cluster granularity. A higher resolution often yields more clusters, but no single value is automatically correct.

Compare several resolutions:

Requires CLI: this example imports the package.

#10

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let base = sc.load("nsc_lc_like")
  |> sc.filter_genes(3)
  |> sc.filter_cells(20, 2500, 5.0)

let trials = sc.sweep(base, [0.2, 0.5, 0.8, 1.2], 100, 15)
println(trials |> map(|x| {resolution: x.resolution, clusters:
x.n_clusters}))
```

The Leiden algorithm was designed to avoid poorly connected communities that can arise with Louvain-style optimization. That guarantee does not make every community a biological cell type.

The one-call workflow

`sc.standard()` makes defaults visible and returns them in `decisions`:

Requires CLI: this example imports the package.

#11

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let raw = sc.load("nsc_lc_like")
let result = sc.standard(raw, 0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0)
println(result.decisions)
```

Use this to begin exploring, then write the explicit pipeline for a final analysis.

What can fool you

- A batch can form a clean cluster.
- A doublet can sit between two cell types.
- UMAP distances and empty spaces are not calibrated biological distances.
- Different PCA signs are mathematically equivalent.
- Numeric cluster IDs are arbitrary and can change between implementations.

Maps, Markers, and Cell Types

Cluster is not cell type

A cluster is an algorithmic community in a selected graph. A cell type is a biological interpretation supported by multiple lines of evidence. One cell type can occupy several states or clusters; one cluster can contain several types.

Visualize the partition

Requires CLI: this example imports the package and writes a file.

#12

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.standard(
  sc.load("nscldc_like"),
  0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0, true
)

write_text("umap.svg", sc.plot_umap(obj, "Teaching populations"))
sc.plot_proportions(obj)
```

The SVG is a view of the result, not the result itself. Keep the parameters, object summary, and cell-level assignments.

Find candidate markers

`marker_table()` compares two clusters:

Requires CLI: this example imports the package.

#13

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.standard(
  sc.load("nsccl_like"),
  0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0, true
)
let markers = sc.marker_table(obj, 0, 1)
|> filter(|x| x.significant)
|> sort(|a, b| if a.log2fc > b.log2fc { -1 } else { 1 })
println(head(table(markers), 12))
```

`log2fc` is positive when the gene is higher in the **first** cluster, matching Seurat's `FindMarkers(ident.1 = 0, ident.2 = 1)` and scanpy's `rank_genes_groups`. Values are on the expression scale, not the log-normalized scale, so they are comparable to a Seurat `avg_log2FC`.

`pct_a` and `pct_b` give the fraction of cells in each cluster with a nonzero value. A gene at `pct_a = 1.0, pct_b = 0.04` is specific; the same fold change at `pct_a = 0.3, pct_b = 0.1` describes a minority of the cluster and is a much weaker identity claim. Means alone hide that difference.

Marker ranking is a hypothesis generator. Review effect size, fraction expressing, specificity across all clusters, known biology, and technical covariates. A tiny p-value with a negligible effect is not a useful identity. This comparison is one cluster against one other; a gene distinguishing 0 from 1 may be shared with cluster 2.

Annotate with positive and negative evidence

For each proposed label, record:

Evidence	Example
Positive markers	Several expected genes are enriched
Negative markers	Incompatible lineage markers are absent
State markers	Stress, cycle, interferon, or activation genes
Reference mapping	Agreement with an appropriate tissue reference
Orthogonal evidence	Protein, morphology, spatial location, or sorting
Uncertainty	Broad label or <code>unknown</code> when evidence conflicts

Do not force every cluster into a precise label. “T cell” can be more honest than “exhausted tissue-resident memory CD8 T cell” when evidence is limited.

Automated references provide a starting point, not ground truth. 10x notes that its reference-based annotation may perform poorly on cancers and cell lines that are not well represented in the reference model.

Visualize marker panels

Requires CLI: this example imports the package and writes files.

#14

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text("top_markers.svg", sc.plot_markers(obj, 5))
write_text(
    "marker_dotplot.svg",
    sc.expr_dotplot(obj, ["MARK0_001", "MARK1_001", "MARK2_001",
    "MARK3_001"])
)
```

On a real immune dataset, use coherent panels rather than one famous marker. Check that marker genes use the same naming convention as the matrix.

Visualization: Every Plot with a Purpose

A plot should answer a stated question. Save the table or cell-level values behind it, because the figure is a view rather than the complete result.

Prepare one object

The examples in this chapter use one clustered object:

Requires CLI: this example imports the package and reads local files.

Requires CLI: this example imports the package.

#15

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.standard(
  sc.load("nsc_lclike"),
  0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0, true
)
```

QC distributions

Before filtering, inspect totals, detected genes, and mitochondrial percentage. The current package provides QC tables; BioLang's general plotting builtins render their columns:

Requires CLI: this example imports the package and reads local files.

Requires CLI: this example imports the package.

#16

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
let with_qc = sc.load("nsc_lcl_like") |> sc.qc()

println("Total counts")
println(hist(col(with_qc.cell_qc_table, "total_counts"), 20))
println("Genes detected")
println(hist(col(with_qc.cell_qc_table, "n_genes"), 20))
println("Mitochondrial percentage")
println(hist(col(with_qc.cell_qc_table, "pct_mito"), 20))
```

These ASCII histograms are useful in logs and remote jobs. For multiple samples, split metrics by sample rather than hiding differences in one pooled distribution.

For a connected SVG inspection surface:

```
write_text("qc-dashboard.svg", sc.plot_qc_dashboard(sc.load("nsc_lcl_like")))
```

`plot_qc_violin()` and `plot_qc_scatter()` provide the two component views when they are needed separately.

PCA plot

Requires CLI: this example imports the package and writes a local file.

```
write_text("pca.svg", sc.plot_pca(obj, "PCA: major linear variation"))
```

Use PCA to inspect dominant linear structure and technical separation. A sample forming its own region can indicate biology, batch, or quality. PCA axes have a global mathematical meaning that UMAP axes do not, but their signs can flip between implementations.

Elbow plot

Requires CLI: this example imports the package.

```
sc.plot_elbow(obj, 15)
```

The ordered ASCII bars show variance explained by each principal component. Look for a gradual leveling rather than pretending there is always one exact cutoff. Check whether later PCs contain coherent biology or mostly noise.

UMAP cluster map

Requires CLI: this example imports the package and writes a local file.

```
write_text("umap.svg", sc.plot_umap(obj, "UMAP by Leiden cluster"))
```

Use UMAP to inspect local neighborhoods, mixing, and outliers. Do not interpret axis values, island area, or long-range distance as calibrated biology. Label the plot with the representation and parameters used.

Use arbitrary per-cell labels for condition, donor, cell type, batch, or phase:

#17

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
    "umap-by-condition.svg",
    sc.plot_embedding(obj, condition_labels, "UMAP by condition")
)
```

Feature plot

Requires CLI: this example imports the package and writes a local file.

#18

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
  "feature-MARK0_001.svg",
  sc.plot_feature(obj, "MARK0_001", "MARK0_001 normalized expression")
)
```

A feature plot colors each UMAP point by one gene's expression. Use several positive and negative markers. A few isolated high cells can be ambient RNA, doublets, or genuine rare expression.

When comparing conditions, keep one colour scale across panels:

#19

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
  "feature-split.svg",
  sc.plot_feature_split(obj, "MARK0_001", condition_labels)
)
```

Violin plot

Requires CLI: this example imports the package and writes a local file.

```
write_text("violin-MARK0_001.svg", sc.plot_violin(obj, "MARK0_001"))
```

The violin compares a gene's normalized expression distribution across clusters. Check both the expressing fraction and magnitude: a broad low signal and a narrow high signal can have similar means.

Marker heatmap

Requires CLI: this example imports the package and writes a local file.

```
write_text("marker-heatmap.svg", sc.plot_markers(obj, 5))
```


The heatmap selects genes with high cluster-vs-rest mean differences and shows mean expression by cluster. It is a compact overview, not a formal replicate-aware differential-expression result.

`plot_group_heatmap()` accepts any per-cell grouping and a chosen gene panel, so the same view can compare cell types, conditions, donors, or cell-cycle phases.

Expression dot plot

Requires CLI: this example imports the package and writes a local file.

#20

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
  "marker-dotplot.svg",
  sc.expr_dotplot(
    obj,
    ["MARK0_001", "MARK1_001", "MARK2_001", "MARK3_001"],
    "Candidate population markers"
  )
)
```

Circle size represents the fraction of cells expressing the gene; color represents mean expression among expressing cells. This separates prevalence from intensity. BioLang's general `dotplot` builtin is a sequence-comparison plot and is unrelated.

Proportion plot

Requires CLI: this example imports the package.

```
sc.plot_proportions(obj)
```

This ordered ASCII chart counts cells per cluster or supplied group. Raw cell fractions can be affected by capture, filtering, and sampling. Perform sample-level compositional analysis before making population claims.

Export all SVG plots

All `plot_*` functions except `plot_elbow` and `plot_proportions` return SVG strings. The latter two print ordered ASCII output for terminals and logs. The advanced gallery and complete export workflow are in Advanced Analysis and Diagnostic Plots.

Every exported figure should be accompanied by:

- the BioLang source and version;
- input identity and filtering summary;
- plot title, groups, genes, and transformations;
- the values or assignments behind the figure;
- a caption stating what the plot can and cannot establish.

Multiple Samples and Batch Effects

Merge first, correct only when needed

Multiple samples are essential for population-level claims. They also introduce technical variation from preparation day, chemistry, lane, sequencing depth, and operator.

Begin by preserving sample identity. Merge objects, inspect them without correction, and ask whether technical factors dominate relevant structure.

Requires CLI: this example imports the package.

#21

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let sample_a = sc.from_matrix(
  matrix([[8, 0, 1], [7, 0, 2], [0, 8, 1]]),
  ["A_MARKER", "B_MARKER", "HOUSEKEEPING"],
  ["A_1", "A_2", "A_3"]
)
let sample_b = sc.from_matrix(
  matrix([[9, 0, 1], [0, 9, 2], [0, 7, 1]]),
  ["A_MARKER", "B_MARKER", "HOUSEKEEPING"],
  ["B_1", "B_2", "B_3"]
)

let merged = sc.merge(sample_a, sample_b, 0, 1)
println(sc.summary(merged))
```

Merge requires aligned genes. Real projects should harmonize feature IDs and genome annotations before this step.

What integration means here

BioLang's current `sc.integrate()` performs deterministic batch centering in PCA space:

Requires CLI: this example imports the package.

#22

▶ CLI Only

Requires CLI — run with: bl run script.bl

```
let prepared = merged
  |> sc.normalize()
  |> sc.variable_genes(3)
  |> sc.run_pca(2)

let corrected = sc.integrate(prepared, [0, 0, 0, 1, 1, 1])
let embedding = corrected.integrated_embedding
println(str(len(embedding)) + " cells x " + str(len(embedding[0])) + "
components")
```

This is lightweight and inspectable. It is not Harmony, fastMNN, Seurat anchors, scVI, or a proof that batches are removed. Use it when a simple location shift is a reasonable model. For complex studies, validate a more appropriate integration method externally or add a tested package.

Do not correct away the question

If all diseased samples are batch A and all controls are batch B, disease and batch cannot be separated computationally. Integration may erase disease or preserve batch, and neither outcome can repair the design.

Evaluate integration by asking:

- Do shared known populations mix across batches?
- Are batch-specific quality problems still visible?
- Are known biological differences retained?
- Do marker patterns remain coherent?
- Are rare populations forced into an unrelated reference?

Composition is also biology

A treatment can change the proportion of a cell type without strongly changing expression inside that type. Count cells per sample and type, but perform inference with a model appropriate for compositional, replicate-level data. Do not run a simple test on pooled cells and call it a patient-level result.

Differential Expression Without Pseudoreplication

Two different marker questions

Cluster marker analysis asks which genes distinguish groups of cells in the current dataset. **Condition differential expression** asks whether expression changes consistently across independent samples.

The first can help annotation. The second supports a population-level claim and requires biological replication.

Why cell-level tests can fail

Cells from one donor are correlated. If each cell is treated as an independent replicate, a method can report extremely small p-values for donor-specific differences. The Squair et al. benchmark showed that methods accounting for biological replication, including pseudobulk approaches, better controlled false discoveries.

Pseudobulk in plain language

For each sample and cell type:

1. return to raw counts;
2. sum counts across cells;
3. create one count profile per biological sample and cell type;
4. use a replicate-aware bulk RNA-seq model with the study design;
5. report effect sizes, uncertainty, and multiple-testing correction.

If eight patients each have monocytes, the monocyte comparison has up to eight independent patient profiles, not thousands of independent monocytes.

Build the table in BioLang

`sc.pseudobulk()` does steps 1 to 3. It sums **raw** counts, not normalized values, within each (cluster, sample) pair. The teaching fixture has no donors, so assign synthetic ones to see the shape:

Requires CLI: this example imports the package and reads local files.

#23

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.standard(
  sc.load("nscldc_like"),
  0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0, true
)

# Stand-in donor labels. In a real study these come from your sample sheet.
let donors = range(0, obj.n_cells) |> map(|i| "donor_" + str(i % 4))

let panel = sc.pseudobulk(obj, donors)
println(colnames(panel))
println(str(nrow(panel)) + " genes x " + str(len(colnames(panel))) + "
profiles")
```

Each column is one independent profile named `<cluster>__<sample>`, and each row is a gene in `obj.genes` order. Four clusters across four donors gives sixteen columns — the real sample size for a condition contrast, against 220 cells.

Counts stay sparse until they are summed, so this does not densify the matrix.

Exploratory paired analysis and the formal boundary

`sc.marker_table()` performs cluster-to-cluster exploratory marker comparison. Its `log2fc` is positive when a gene is higher in the first cluster, matching Seurat's `FindMarkers` and scanpy's `rank_genes_groups`. It is not a replacement for a pseudobulk negative-binomial model such as those used by edgeR or DESeq2.

BioLang includes `sc.pseudobulk_profiles()` and `sc.paired_pseudobulk_de()` for a transparent donor-paired log2-CPM analysis. It also provides volcano, MA, paired-donor, and pseudobulk-PCA plots. This is appropriate for teaching, exploration, and cross-checking effect direction.

Formal count inference remains an explicit handoff. Write the raw panel out and invoke a validated R, Python, container, or workflow step using DESeq2, edgeR, or another suitable negative-binomial method. For paired samples, represent the pairing in the design, for example `~ donor + condition`. Keep this boundary explicit in the report.

A project handoff might look like:

```
results/pseudobulk_counts.tsv
results/sample_metadata.tsv
scripts/run_deseq2.R
results/differential_expression.tsv
validation/session_info.txt
```

Questions for every DE table

- What is the independent biological unit?
- How many units are in each condition?
- Was pairing represented?
- Were batches and major covariates included?
- Were raw counts used by a suitable count model?
- Was the cell type defined without using the tested condition in a circular way?
- Are effect sizes meaningful, not merely significant?
- Were results stable to QC and annotation sensitivity analyses?

For clinicians

A differentially expressed gene is an association in a specified dataset. It does not establish causality, drug response, prognosis, or diagnostic accuracy. Those require separate study designs and validation.

Advanced Analysis and Diagnostic Plots

Advanced analysis does not mean applying more algorithms. It means testing a biological question at the correct level, showing where the result came from, and checking whether reasonable analysis choices change the conclusion.

The complete runnable example for this chapter is distributed with the package:

```
bl examples singlecell --copy singlecell-examples
cd singlecell-examples
bl run advanced_analysis.bl
```

It creates `singlecell-results/` with SVG figures, differential-expression tables, composition tests, and ranked genes. The fixture has four donors measured in control and treated conditions, with T and B cells in every sample. The code excerpts below use `obj`, `analysed`, `donors`, `conditions`, and `cell_types` exactly as they are defined in that complete example.

Inspect quality as a connected problem

#24

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
  "singlecell-results/qc-dashboard.svg",
  sc.plot_qc_dashboard(obj)
)
```

The dashboard combines library-size, detected-gene, and mitochondrial distributions with the counts-versus-genes relationship. A threshold should be chosen from these distributions, sample context, and protocol knowledge. It should not be copied blindly from another tissue.

Label an embedding by the question

Clusters are only one way to colour an embedding. Compare condition, donor, curated cell type, batch, and cell-cycle phase:

#25

▶ CLI Only

Requires CLI — run with: bl run script.bl

```

write_text(
  "singlecell-results/umap-condition.svg",
  sc.plot_embedding(analysed, conditions, "UMAP by treatment")
)

write_text(
  "singlecell-results/feature-split.svg",
  sc.plot_feature_split(
    analysed, "IFIT1", conditions, "IFIT1 by treatment"
  )
)

```

The split feature plot uses the same expression scale in every panel. Separate plots with independently chosen colour limits can manufacture an apparent difference.

For gene panels grouped by cell type, condition, donor, or phase:

#26

▶ CLI Only

Requires CLI — run with: bl run script.bl

```

write_text(
  "singlecell-results/group-heatmap.svg",
  sc.plot_group_heatmap(
    analysed,
    ["IFIT1", "MS4A1", "CD3D"],
    cell_types,
    "Lineage and response genes"
  )
)

```

Build donor-level profiles

#27

▶ Run

▶▶ Run All Above + This

```

let profiles = sc.pseudobulk_profiles(
  obj, donors, conditions, cell_types, 5
)

println(profiles.profiles)
println(
  str(len(profiles.profiles)) + " profiles x " +
  str(len(profiles.genes)) + " genes"
)

```

Raw counts are summed for each donor x condition x cell-type combination. Profiles with fewer than five cells are excluded. The returned record contains:

- `counts` : genes x pseudobulk profiles, preserving raw sums;
- `logcpm_matrix` : profiles x genes for transparent exploration and PCA;
- `profiles` : donor, condition, cell type, cell count, and library size;
- `genes` : row identifiers corresponding to the matrices.

Run an explicit paired exploratory test

#28 ▶ Run ▶▶ Run All Above + This

```
let de = sc.paired_pseudobulk_de(
  obj,
  donors,
  conditions,
  cell_types,
  "T",
  "control",
  "treated",
  5
)

let top = de
|> sort(|a, b| if a.padj < b.padj { -1 } else { 1 })
|> take(10)
println(table(top))
```

This function pairs each donor's control and treated log2-CPM profile, applies a paired test per gene, and performs Benjamini-Hochberg correction. It is useful for transparent exploration and teaching.

It is not a replacement for a publication-grade negative-binomial model. For formal inference, export `profiles.counts` and use DESeq2, edgeR, or another validated count model with an explicit design such as `~ donor + condition`. Include batch and clinical covariates when the study design requires them.

Use three complementary DE plots

#29 ▶ CLI Only Requires CLI — run with: bl run script.bl

```

write_text("singlecell-results/volcano.svg", sc.plot_volcano(de))
write_text("singlecell-results/ma.svg", sc.plot_ma(de))
write_text(
  "singlecell-results/paired-donors.svg",
  sc.plot_donor_expression(
    obj, "IFIT1", donors, conditions, cell_types,
    "T", "control", "treated", 5
  )
)

```

- The **volcano plot** combines effect magnitude and adjusted significance.
- The **MA plot** reveals expression-dependent fold-change behaviour.
- The **paired-donor plot** shows whether the direction is shared by donors or driven by one sample.

A convincing volcano point with inconsistent donor trajectories deserves investigation, not a stronger claim.

Check samples in pseudobulk PCA

#30

▶ CLI Only

Requires CLI — run with: bl run script.bl

```

write_text(
  "singlecell-results/pseudobulk-pca.svg",
  sc.plot_pseudobulk_pca(
    obj, donors, conditions, cell_types, "T", 5
  )
)

```

Each point is now a biological sample, not a cell. Colour shows condition and labels identify donors. Inspect outliers, pairing, and whether treatment separation is larger than donor-to-donor variation.

Test cell-type composition at sample level

#31

▶ CLI Only

Requires CLI — run with: bl run script.bl

```

let composition = sc.composition_table(
  donors, conditions, cell_types
)
let composition_tests = sc.paired_composition_test(
  donors, conditions, cell_types,
  "control", "treated", 10
)

write_text(
  "singlecell-results/composition.svg",
  sc.plot_composition(donors, conditions, cell_types)
)
println(table(composition_tests))

```

The table includes zero-count cell types so missing populations do not disappear from the analysis. The paired test treats donors, rather than cells, as replicates. Cell fractions are compositional and affected by capture and QC; specialized compositional models remain preferable for complex studies.

Check clustering quality and sensitivity

#32

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```

let diagnostics = sc.cluster_diagnostics(analysed)
let stability = sc.cluster_stability(
  analysed, [0.2, 0.4, 0.6, 0.8, 1.0], 8
)

write_text(
  "singlecell-results/silhouette.svg",
  sc.plot_silhouette(diagnostics)
)
write_text(
  "singlecell-results/cluster-stability.svg",
  sc.plot_cluster_stability(stability)
)

```

Silhouette coefficients describe separation in the PCA or integrated embedding. The resolution sweep reports cluster count, mean silhouette, and adjusted Rand index against the previous resolution. No single score selects a biologically correct resolution: combine diagnostics with marker coherence, replication, and the question being asked.

Rank genes and test pathways

#33

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
let ranked = sc.ranked_genes(de)
let sets = {
  interferon_response: ["IFIT1", "ISG15", "MX1", "OAS1"],
  antigen_presentation: ["HLA-DRA", "HLA-DPA1", "CD74"]
}
let enrichment = gsea(ranked, sets)

write_text(
  "singlecell-results/enrichment.svg",
  sc.plot_enrichment(enrichment)
)
```

`ranked_genes()` uses the signed effect and p-value to create the `gene, score` table expected by BioLang's `gsea()` builtin. Gene identifiers, the tested background, and gene-set versions must be recorded.

Compare methods without a Venn diagram

#34

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
write_text(
  "singlecell-results/de-overlap.svg",
  sc.plot_de_overlap(
    [paired_results, deseq2_results, edger_results],
    ["BioLang paired", "DESeq2", "edgeR"]
  )
)
```

The UpSet-style plot scales to more result sets than a Venn diagram. Agreement is useful, but inspect effect directions and ranks as well as set membership.

Minimum advanced report

An advanced report should contain:

1. sample and donor counts before and after QC;
2. QC distributions split by sample;
3. embeddings coloured by sample, condition, and annotation;
4. pseudobulk sample PCA;

5. the statistical design and replicate definition;
6. volcano, MA, and paired-donor views;
7. composition results at sample level;
8. clustering sensitivity diagnostics;
9. enrichment with gene-set provenance;
10. R or Python validation and session versions.

Trajectories, Cell Cycle, and Doublets

Cell type and cell state

A type describes relatively stable identity; a state describes a more temporary program such as activation, stress, proliferation, or response to interferon. State programs can appear across several types and can dominate clustering.

Score a gene program

BioLang can average a selected gene set per cell:

Requires CLI: this example imports the package.

#35

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
import "singlecell" as sc

let obj = sc.standard(
  sc.load("nscic_like"),
  0.5, 100, 15, 20, 2500, 5.0, 3, 10000.0, true
)
let marker_indices = obj.hvg
  |> take(8)
let scored = sc.gene_module_score(obj, marker_indices)
println(take(scored.module_scores, 10))
```

A module score depends on gene-set quality and the matrix used. It is not an assay measurement or pathway activation proof.

Cell cycle

`sc.cell_cycle()` accepts S-phase and G2/M gene lists and returns per-cell scores and phases. Use organism-appropriate genes. Decide whether cycle is a nuisance, the biological question, or both before regressing or filtering it. Removing a genuine proliferating population would erase biology.

Pseudotime

Pseudotime orders cells along a graph from a chosen start cell:

Requires CLI: this example imports the package.

#36

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
let ordered = obj |> sc.pseudotime(0)
println(take(ordered.pseudotime, 12) |> map(|v| round(v, 3)))
println(take(sc.order_by_pseudotime(ordered).barcodes, 6))
println(take(sc.pseudotime_bins(ordered.pseudotime, 6), 12))
```

`order_by_pseudotime` returns a new object whose cells are sorted, not a list of positions, so read its `barcodes` to see the order.

The result is a relative graph distance, not clock time. Changing the root, neighbors, or included cells can change the order. A branch may represent different lineages, activation, cell cycle, or an artifact.

Use trajectory analysis when biology plausibly contains a continuum and when intermediate cells were sampled. Support direction with known markers, experimental time points, lineage tracing, RNA velocity, or another independent source. Do not infer causality from an arrow drawn on UMAP.

Doublets as false intermediates

Doublets often express markers from two lineages and appear between them. This can resemble a transitional state. Before naming a bridge population:

- inspect doublet scores;
- check total counts and detected genes;
- examine incompatible lineage markers;
- compare samples and expected loading rate;
- seek orthogonal evidence.

The honest label `uncertain` is preferable to a memorable but unsupported new cell state.

A Tumor Microenvironment Case Study

The question

Imagine a research team asking: “Can we recover the major populations in a small tumor-like count matrix, and are the clusters stable enough to support marker review?”

This is an exploratory engineering and analysis question. The synthetic fixture has known truth, so it can test workflow plumbing. It cannot validate a clinical claim.

Complete script

Save this as `singlecell_book_case_study.bl` in the `packages` directory after generating `nsclc_like`.

Requires CLI: this complete example imports the package and reads/writes local files.

#37

 CLI OnlyRequires CLI — run with: `bl run script.bl`

```

import "singlecell" as sc

let raw = sc.load("nscclc_like")
println("raw: " + str(raw.n_cells) + " cells x " + str(raw.n_genes) + "
genes")

let obj = raw
  |> sc.filter_genes(3)
  |> sc.filter_cells(20, 2500, 5.0)
  |> sc.normalize(10000.0)
  |> sc.variable_genes(100)
  |> sc.run_pca(20)
  |> sc.neighbors(15)
  |> sc.cluster_leiden(15, 0.5)

println("kept: " + str(obj.n_cells) + " cells")
println("clusters: " + str(obj.clusters |> unique |> len))
println(sc.summary(obj))

sc.plot_elbow(obj, 15)
sc.plot_proportions(obj)

let truth_rows = read_lines("nscclc_like/truth.csv")
  |> drop(1)
  |> filter(|line| len(line) > 0)
  |> map(|line| split(line, ","))
let truth_barcodes = truth_rows |> map(|row| row[0])
let truth_labels = truth_rows |> map(|row| int(row[1]))
let expected = obj.barcodes
  |> map(|barcode| truth_labels[find_index(truth_barcodes, |x| x ==
barcode)])

println("ARI vs fixture truth: " + str(round(ari(obj.clusters, expected),
4)))

write_text("case-study-umap.svg", sc.plot_umap(obj, "Tumor-like teaching
data"))
write_csv(
  table(range(0, obj.n_cells) |> map(|i| {
    barcode: obj.barcodes[i],
    cluster: obj.clusters[i],
    truth: expected[i]
  })),
  "case-study-labels.csv"
)

```

Expected interpretation

With the current implementation and fixture, 220 cells remain, four clusters are found, and adjusted Rand index (ARI) against the known partition is 1.0000. ARI compares partitions while ignoring arbitrary numeric cluster names.

This means the seeded populations are cleanly recovered. It does not mean:

- every real tumor separates into four populations;
- resolution `0.5` is universally correct;

- the marker simulation resembles every technology;
- BioLang, Scanpy, and Seurat are identical for all datasets;
- the result says anything about a real patient.

Review it as four professionals

Biologist: Are marker blocks coherent, and what real lineages would require positive and negative evidence?

Researcher: Would alternate QC thresholds, `k`, and resolution retain the partition?

Programmer: Does sparse storage remain bounded, and where does exact kNN become expensive?

Clinician: What additional cohort, assay, endpoint, and governance would be required before this could affect care?

Extend the case study

Run resolutions `0.2`, `0.5`, `0.8`, and `1.2`. Compare cluster counts, ARI, marker coherence, and whether small clusters contain high-mitochondrial or high-doublet-score cells. Record all outcomes rather than retaining only the most attractive map.

Validate with Scanpy and Seurat

What validation can establish

Cross-library validation can detect mistakes in orientation, filtering, barcodes, graph construction, and label export. It cannot prove that one workflow is biologically correct merely because another package agrees.

The package includes independent reference scripts. Copy its examples and enter the resulting working directory:

```
bl examples singlecell --copy singlecell-examples
cd singlecell-examples

validation/
  biolang_reference.bl
  scanpy_reference.py
  seurat_reference.R
  compare_labels.py
```

Run the three workflows

Create the isolated Scanpy environment once in the copied example directory:

```
# Windows
py -3.12 -m venv .venv-scanpy
.venv-scanpy\Scripts\python -m pip install -r validation\requirements-
scanpy.txt

# macOS/Linux
python3.12 -m venv .venv-scanpy
.venv-scanpy/bin/python -m pip install -r validation/requirements-scanpy.txt
```

Then run from `singlecell-examples`:

```
.venv-scanpy\Scripts\python make_demo_10x.py --output validation_data
bl run validation/biolang_reference.bl
.venv-scanpy\Scripts\python validation/scanpy_reference.py validation_data
validation_scanpy_labels.csv
Rscript validation/seurat_reference.R validation_data
validation_seurat_labels.csv
```

The generated matrix is `<your working directory>/validation_data/`. Every relative data and label path in this section is resolved from the copied example directory.

The commands above show Windows paths. On macOS/Linux, use `.venv-scanpy/bin/python` and `bl`. On Windows, use the full `Rscript.exe` path if it is not on `PATH`.

Compare by barcode and ARI:

```
.venv-scanpy\Scripts\python validation/compare_labels.py
validation_data/truth.csv validation_biolang_labels.csv
.venv-scanpy\Scripts\python validation/compare_labels.py
validation_scanpy_labels.csv validation_biolang_labels.csv
.venv-scanpy\Scripts\python validation/compare_labels.py
validation_seurat_labels.csv validation_biolang_labels.csv
```

Verified result

The seeded fixture was checked on 2026-07-27:

Runtime	Key versions	Cells	Clusters	ARI vs BioLang
BioLang	repository sparse workflow	220	4	1.0000 vs truth
Scanpy	Python 3.12.10; Scanpy 1.12.3; Numba 0.65.1; igraph 1.0.0	220	4	1.0000
Seurat	5.5.1; Matrix 1.7.4	220	4	1.0000

Cluster numbers need not match. ARI compares which cells are grouped together. PCA signs can also differ without changing the geometry.

The current official workflows remain useful comparators: Scanpy preprocessing and clustering and the Seurat guided clustering tutorial.

Validate more than one number

For a real dataset compare:

- retained cells and genes by stable identifier;

- per-cell totals, genes detected, and mitochondrial percentage;
- normalized value summaries;
- HVG overlap, allowing for method differences;
- PCA variance explained and pairwise structure;
- graph degree and connected components;
- cluster ARI or NMI;
- marker effect directions and ranks;
- exported file dimensions and checksums.

Set acceptance criteria before seeing the result. Investigate disagreement instead of tuning until the tools match.

Validate donor-aware differential expression

After copying the examples, run:

```
bl run advanced_analysis.bl
python validation/advanced_reference.py
Rscript validation/advanced_reference.R
```

On Windows, use the full `Rscript.exe` path when necessary. The scripts independently aggregate the same paired donor fixture, calculate log2 CPM, run paired tests, and compare BioLang's IFIT1 effect and p-value.

The reference check on 2026-07-27 agreed to numerical tolerance with Python 3.13.14, NumPy 2.4.6, SciPy 1.18.0, and R 4.6.1:

```
IFIT1 log2fc = 1.701812
paired p     = 0.000242968
```

This validates implementation parity for the stated exploratory model. It does not turn a paired t-test on log2 CPM into DESeq2. A real study should separately validate raw pseudobulk counts, sample design, contrasts, dispersion estimates, effect directions, adjusted p-values, and gene identifiers.

Scale, Reproduce, and Report

Know where memory goes

Real count matrices are mostly zero. BioLang keeps 10x counts and normalized matrices sparse and computes PCA without materializing a dense centered matrix. This is important because a dense matrix of one million cells by 30,000 genes is not a practical laptop object.

Current exact nearest-neighbor search is quadratic in cell count. The graph itself is sparse, but building exact neighbors becomes the practical limit for large atlases. Use a representative subset, a validated approximate-neighbor tool, or remote compute until approximate indexing is available in BioLang.

Checkpoint intermediate results

Expensive stages should produce named, checksummed artifacts:

```
results/  
  01-qc-summary.tsv  
  02-filtered.zarr/  
  03-pca-summary.tsv  
  04-cell-clusters.tsv  
  05-marker-review.tsv  
figures/  
logs/
```

BioLang reads and writes sparse AnnData Zarr **x** plus observation and variable index names. Arbitrary AnnData metadata columns and auxiliary layers are not yet fully preserved. Direct **.h5ad** I/O requires external conversion. Verify a roundtrip before relying on it for archival interchange.

Record the environment

At minimum save:

- BioLang version and commit;
- package source and version;

- operating system and architecture;
- command line;
- input accession, path, size, and checksum;
- genome and annotation version;
- every non-default parameter;
- Python/R/container versions used for validation;
- random seeds where relevant;
- warnings and failed attempts.

Separate exploration from final analysis

Exploration is allowed to be iterative. The final workflow should be rerunnable from immutable inputs and should not depend on clicking cells or manually editing a result table without an audit trail.

A literate `.b1n` notebook can explain decisions and show compact outputs. Keep the production pipeline in scripts when it needs batch execution, retries, and large artifacts.

Remote execution

Large projects may execute on a workstation, scheduler, container platform, or remote service. Preserve the same logical contract:

```
source + parameters + input identities -> logs + status + artifacts
```

Do not silently depend on local files that a remote worker cannot access. Stage inputs explicitly, avoid embedding credentials in scripts, and verify artifacts after transfer.

Report decisions, not only pictures

A useful methods section states:

- how cells and genes were filtered;
- how counts were normalized;
- how HVGs and PCs were selected;

- graph and clustering parameters;
- how labels were assigned and reviewed;
- how sample replication entered statistical tests;
- which sensitivity analyses were performed;
- which steps used external tools.

Failure Modes and Review Checklist

Common attractive mistakes

“The UMAP has islands, so these are cell types”

UMAP is a visualization of a selected representation. Inspect markers, technical covariates, graph stability, and reference evidence.

“There are 10,000 cells, so the study is well replicated”

Count independent donors or experimental units. Cell count and replicate count answer different power questions.

“Batch correction mixed the samples, so integration worked”

Perfect mixing can erase biology. Check known controls, marker preservation, and whether condition is confounded with batch.

“The smallest cluster is a rare discovery”

It can also be low quality, doublets, ambient RNA, a sample-specific artifact, or overclustering. Require stronger evidence for rarer claims.

“A famous marker appeared, so the label is certain”

Use panels, negative evidence, tissue context, references, and uncertainty.

“Scanpy and Seurat agree, so the biology is proven”

Agreement validates implementation behavior on the tested inputs. The tools can share assumptions and biases.

Technical review

- ☐ Input dimensions and orientation are known.
- ☐ Stable gene IDs and symbols are preserved.
- ☐ Raw counts are retained.
- ☐ Sample and donor IDs are attached before merging.
- ☐ QC distributions were inspected per sample.
- ☐ Cells and genes removed at each step are counted.
- ☐ Sparse matrices were not accidentally densified.
- ☐ PCA, **k**, and resolution choices are recorded.
- ☐ Numeric cluster IDs are not treated as biological labels.
- ☐ Outputs join by barcode, not row position alone.
- ☐ Random seeds and software versions are recorded.

Scientific review

- ☐ The question and experimental unit are explicit.
- ☐ Replicates, pairing, and confounders are represented.
- ☐ Marker annotation uses positive and negative evidence.
- ☐ Doublets and low-quality clusters were considered.
- ☐ Condition DE uses replicate-aware inference.
- ☐ Sensitivity analyses cover reasonable QC and clustering choices.
- ☐ Claims distinguish association, mechanism, and prediction.
- ☐ Independent or orthogonal validation is identified.

Clinical review

- ☐ The result is labeled discovery, validation, or clinical use.
- ☐ The population matches the intended use population.
- ☐ Performance and uncertainty were measured on held-out data.
- ☐ Privacy, consent, and data access are appropriate.
- ☐ A qualified human reviews any decision-affecting output.
- ☐ No exploratory cluster is presented as a diagnostic category.

Final question

Could another team recover the same result, understand every major decision, and identify what evidence would change the conclusion? If not, the analysis is not finished.

BioLang Single-Cell API

Import the package with:

Requires CLI: package imports are not available in the browser runner.

```
import "singlecell" as sc
```

BioLang resolves this name through the package's `[lib].entry` declaration in `biolang.toml`.

Object creation and access

Function	Purpose
<code>sc.load(path)</code>	Read a 10x MEX directory into a sparse object
<code>sc.from_matrix(matrix, genes, barcodes)</code>	Build an object from a dense cells-by-genes matrix
<code>sc.summary(obj)</code>	Return dimensions and pipeline-stage flags
<code>sc.n_cells</code> , <code>sc.n_genes</code>	Return current dimensions
<code>sc.get_matrix</code>	Return raw counts
<code>sc.get_norm_matrix</code>	Return normalized values
<code>sc.get_genes</code> , <code>sc.get_barcodes</code>	Return identifiers
<code>sc.get_hvg_genes</code>	Return selected HVG names
<code>sc.get_clusters</code>	Return cluster labels
<code>sc.get_pseudotime</code>	Return per-cell pseudotime
<code>sc.get_doublet_scores</code>	Return per-cell scores

QC and preprocessing

Function	Purpose
<code>sc.qc(obj)</code>	Add cell and gene QC tables
<code>sc.filter_cells(obj, min_genes, max_genes, max_pct_mito)</code>	Subset cells
<code>sc.filter_genes(obj, min_cells)</code>	Subset genes
<code>sc.normalize(obj, target)</code>	Total-count normalization plus log1p
<code>sc.sctransform(obj)</code>	Regularized negative-binomial residual transform
<code>sc.variable_genes(obj, n)</code>	Select genes by dispersion
<code>sc.scale(obj)</code>	Dense mean-center and standardize; raises on sparse objects
<code>sc.run_pca(obj, n_pcs)</code>	Sparse-aware PCA
<code>sc.neighbors(obj, k)</code>	Exact kNN graph on PCA or integrated embedding
<code>sc.cluster(obj, k)</code>	k-means alternative
<code>sc.cluster_leiden(obj, k, resolution)</code>	Leiden clustering on stored graph
<code>sc.standard(...)</code>	Run and report the standard workflow
<code>sc.sweep(obj, resolutions, n_hvg, k)</code>	Compare clustering resolutions

Annotation and state

Function	Purpose
<code>sc.marker_table(obj, a, b)</code>	Exploratory markers for cluster A vs B; <code>log2fc > 0</code> means higher in A

Function	Purpose
<code>sc.pseudobulk(obj, sample_ids)</code>	Sum raw counts per (cluster, sample) for replicate-aware DE
<code>sc.pseudobulk_profiles(obj, donors, conditions, cell_types, min_cells)</code>	Build raw and log-CPM donor-condition-cell-type profiles
<code>sc.paired_pseudobulk_de(...)</code>	Exploratory paired donor-level condition test
<code>sc.donor_gene_table(...)</code>	Return per-donor values for one gene and two conditions
<code>sc.composition_table(...)</code>	Return donor-condition cell-type fractions, including zeros
<code>sc.paired_composition_test(...)</code>	Test paired donor-level composition changes
<code>sc.cluster_diagnostics(obj)</code>	Calculate per-cell and mean silhouette coefficients
<code>sc.cluster_stability(obj, resolutions, k)</code>	Sweep Leiden resolution with silhouette and ARI
<code>sc.ranked_genes(de_results)</code>	Prepare signed <code>gene</code> , <code>score</code> input for <code>gsea()</code>
<code>sc.cell_cycle(obj, s_genes, g2m_genes)</code>	Score cell-cycle phase
<code>sc.gene_module_score(obj, gene_indices)</code>	Average selected genes per cell
<code>sc.doublets(obj, n_sim)</code>	Simulate mixtures and score doublets
<code>sc.flag_doublets(obj, threshold)</code>	Add boolean doublet flags

Multiple samples and trajectories

Function	Purpose
<code>sc.merge(a, b, batch_a, batch_b)</code>	Merge gene-aligned objects
<code>sc.integrate(obj, batch_ids)</code>	Center batches in PCA space
<code>sc.integrate_from_column(obj, batch_col)</code>	Integrate from a top-level field such as <code>batch_ids</code>
<code>sc.pseudotime(obj, start_cell)</code>	Graph distance from a root cell
<code>sc.order_by_pseudotime(obj)</code>	Return cell order
<code>sc.pseudotime_bins(pt, n_bins)</code>	Assign relative bins
<code>sc.smooth_along_pseudotime(obj, gene_idx, n_bins)</code>	Bin gene expression along order

Visualization

Function	Return
<code>sc.plot_umap(obj, title)</code>	SVG string
<code>sc.plot_pca(obj, title)</code>	SVG string
<code>sc.plot_feature(obj, gene, title)</code>	SVG string
<code>sc.plot_violin(obj, gene)</code>	SVG string
<code>sc.plot_markers(obj, n)</code>	SVG string
<code>sc.expr_dotplot(obj, genes, title)</code>	SVG string
<code>sc.plot_elbow(obj, n_pcs)</code>	Printed ASCII chart
<code>sc.plot_proportions(obj, groups)</code>	Printed ASCII chart
<code>sc.plot_qc_violin(obj, title)</code>	SVG string
<code>sc.plot_qc_scatter(obj, title)</code>	SVG string
<code>sc.plot_qc_dashboard(obj, title)</code>	SVG string
<code>sc.plot_embedding(obj, groups, title)</code>	SVG string

Function	Return
<code>sc.plot_feature_split(obj, gene, groups, title)</code>	SVG string
<code>sc.plot_group_heatmap(obj, genes, groups, title)</code>	SVG string
<code>sc.plot_volcano(results, title, fc, padj)</code>	SVG string
<code>sc.plot_ma(results, title, fc, padj)</code>	SVG string
<code>sc.plot_donor_expression(...)</code>	SVG string
<code>sc.plot_pseudobulk_pca(...)</code>	SVG string
<code>sc.plot_composition(...)</code>	SVG string
<code>sc.plot_de_overlap(result_sets, names, title)</code>	SVG string
<code>sc.plot_enrichment(results, title)</code>	SVG string
<code>sc.plot_cluster_stability(stability, title)</code>	SVG string
<code>sc.plot_silhouette(diagnostics, title)</code>	SVG string

Important boundaries

- Exact kNN is currently quadratic in cell count.
- `scale()` is dense-only and raises on an object loaded from a 10x directory, because centering would materialize every zero. Use `run_pca()`.
- HVG ranking is global CV² with no binning by mean expression, so it favors low-expression genes more than Scanpy's `seurat` flavor or Seurat VST.
- Integration is PCA batch centering, not a full atlas-mapping method.
- `marker_table()` is exploratory and cell-level. It is not replicate-aware condition DE — use `pseudobulk()` and a count model for that.
- `paired_pseudobulk_de()` uses a paired test on log2 CPM for transparent exploration. Publication-grade count inference should use the raw pseudobulk profiles with a validated negative-binomial model and explicit study design.
- `sctransform()` returns a dense matrix; Pearson residuals are nonzero where counts were zero, so there is no sparse result to preserve.
- AnnData Zarr metadata interchange is currently limited.

File Formats and Conversion

10x Matrix Exchange

Expected files:

```
matrix.mtx or matrix.mtx.gz  
features.tsv or features.tsv.gz  
barcodes.tsv or barcodes.tsv.gz
```

BioLang reads these natively with `sc.load(path)`. Confirm whether the directory is raw or filtered and whether it contains gene expression alone or additional feature types.

AnnData Zarr

BioLang can read and write sparse AnnData Zarr stores. The current native interchange preserves:

- sparse `X`;
- observation index names;
- variable index names.

It does not yet promise lossless transfer of arbitrary `obs` / `var` columns, layers, embeddings, graphs, or unstructured metadata. Roundtrip a representative object and compare dimensions, names, nonzero values, and checksums before using it as an archive.

H5AD

Direct `.h5ad` I/O is not built into the runtime. Convert with Python/anndata:

```
import anndata as ad

adata = ad.read_h5ad("input.h5ad")
adata.write_zarr("input.zarr")
```

Then read the Zarr store through the BioLang AnnData functions. Record the anndata version and conversion command.

Seurat objects

An `.rds` Seurat object is R-specific and can contain multiple assays, layers, reductions, and graphs. Export the exact matrix and metadata required for the handoff, preferably with stable gene and barcode keys. Never assume row order alone survived conversion.

Delimited tables

CSV and TSV are useful for compact metadata and results. Note the argument order: the table comes first, the path second.

Requires CLI: this example writes a local file.

#38

▶ CLI Only

Requires CLI — run with: `bl run script.bl`

```
let labels = table([
  {barcode: "cell_1", sample: "S01", cluster: 0},
  {barcode: "cell_2", sample: "S01", cluster: 1}
])
write_csv(labels, "cell-labels.csv")
```

Join tables by barcode and sample ID. Barcodes are not necessarily globally unique across samples.

Large and private inputs

Do not put protected patient matrices in a public package or book. For public datasets, record accession, expected files, checksums, and license. For private datasets, use approved storage and access controls, and keep a manifest that can be audited without exposing the data.

Glossary

Adjusted Rand index (ARI): Agreement between two partitions, corrected for chance and independent of numeric cluster names.

Ambient RNA: RNA released into the suspension and captured by droplets that may not contain the source cell.

Barcode: Sequence used to associate molecules with a captured droplet or cell library.

Batch: A group sharing a technical processing event. Batch can be confounded with biology.

Cell state: A relatively temporary program such as activation, stress, or cell cycle.

Cell type: A biological identity supported by lineage, function, markers, and context. It is not identical to an algorithmic cluster.

Cluster: A group produced by an algorithm under selected data and parameters.

Count matrix: Table of observed molecule counts for cells and genes.

Differential expression: A statistical comparison of expression between defined groups under a specified design.

Doublet: One barcode representing two captured cells.

Experimental unit: The smallest independently assigned or sampled unit that supports inference, often a donor, animal, organoid, or culture.

Feature: A measured item, usually a gene but potentially an antibody tag or guide.

Gene: A genomic region that contributes to a functional RNA or protein product.

Highly variable gene (HVG): Gene selected because its variation is useful for describing cell structure under a particular method.

Integration: Construction of a representation intended to align shared biology across samples or batches.

Library size: Total observed counts for one cell library.

Marker: Gene whose expression helps distinguish a group. A marker is context-dependent and not necessarily unique.

Mitochondrial fraction: Fraction of a cell's counts assigned to mitochondrial genes.

Neighbor graph: Graph connecting each cell to similar cells in a selected representation.

Normalization: Transformation intended to make measurements more comparable for a downstream purpose.

PCA: Principal component analysis, a linear low-dimensional representation.

Pseudobulk: Counts aggregated across cells within each biological sample and cell type for replicate-aware analysis.

Pseudoreplication: Treating correlated observations, such as cells from one donor, as independent experimental replicates.

Pseudotime: Relative ordering along an inferred cellular continuum. It is not clock time.

RNA: Molecules involved in gene expression and regulation. Messenger RNA is used as a proxy for transcriptional activity.

Sparse matrix: Matrix stored using nonzero entries rather than every zero.

Transcript: RNA molecule produced from a gene.

UMAP: Nonlinear low-dimensional visualization. Its visual distances are not calibrated biological distances.

UMI: Unique molecular identifier used to reduce duplicate counting of the same captured molecule.

Exercises and Answers

Exercises

1. Experimental units

Four mice per condition each contribute 3,000 cells. What is the sample size for a condition comparison? What does the large cell count improve?

2. Matrix orientation

A 10x Matrix Market header reports 20,000 rows and 5,000 columns. What do those usually represent? What orientation does `sc.load()` expose?

3. QC sensitivity

Run the teaching workflow with maximum mitochondrial percentages of `3.0`, `5.0`, and `10.0`. Record retained cells, clusters, and ARI.

4. Resolution sensitivity

Use `sc.sweep()` across `[0.2, 0.5, 0.8, 1.2]`. Which result has the most clusters? Does more detail automatically mean more truth?

5. Annotation

For one real cluster, prepare a table of positive markers, negative markers, state programs, reference evidence, and uncertainty. Which label is the narrowest one the evidence supports?

6. Validation

Why can BioLang cluster `0` correspond to Scanpy cluster `2` while ARI remains 1.0?

7. Differential expression

Write a pseudobulk table design for six patients measured before and after treatment. What are the rows, columns, pairing variable, and primary contrast?

8. Clinical claim

Rewrite “Cluster 4 proves this patient will respond to therapy” as an accurate exploratory statement.

Answers

1

The independent sample size is four mice per condition. More cells improve the description of within-mouse heterogeneity and rare populations, not the number of independently treated mice.

2

Rows usually represent genes/features and columns represent barcodes/cells.

`sc.load()` exposes a cells-by-genes matrix.

3

The exact numbers are empirical. The important answer is the recorded change and whether removed cells concentrate in a population. A threshold is not justified because it gives the preferred ARI.

4

Higher resolution generally creates more clusters. More clusters can split states or technical variation and does not automatically improve biological truth.

5

The answer must use multiple genes and conflicting evidence. `unknown` or a broad lineage is acceptable when evidence is insufficient.

6

Cluster IDs are arbitrary labels. ARI compares co-membership: which cells are grouped together.

7

Rows are genes and columns are patient-by-timepoint pseudobulk profiles within a specified cell type. Metadata contains patient and timepoint; the model uses patient pairing and tests the post-vs-pre contrast.

8

“In this exploratory dataset, a cell population labeled cluster 4 is associated with response and requires validation in independent patients before predictive or clinical interpretation.”

References

Workflows and data formats

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- Scanpy. Preprocessing and clustering tutorial
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How to use these references

Documentation links describe current software behavior and can change with versions. Papers describe methods and evidence at publication time. Record the

software version actually used, read method-specific documentation, and do not substitute this book for protocol, statistical, clinical, privacy, or regulatory review.